Kaduna.lk - A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Vehicle maintenance in Sri Lanka is predominantly reactive or based on fixed manufacturer intervals, neither of which considers actual component condition or driving behavior. This leads to unexpected breakdowns, increased repair costs, and safety risks. Mechanical failures contribute to approximately 10–15% of road accidents globally, while unplanned breakdowns increase fleet maintenance costs by up to 20%. This research proposes a Predictive Vehicle Maintenance System that estimates the health of critical components—tires, brake pads, engine oil, and batteries—and predicts their Remaining Useful Life (RUL). Since standard OBD-II does not provide direct readings for brake pad thickness, tire tread, or oil quality, the system infers component degradation from indirect OBD-II signals (braking patterns, engine load, temperature profiles) combined with driver behavior data from smartphone sensors and historical service records. Data is collected via an ELM327 Bluetooth OBD-II adapter and smartphone sensors (accelerometer, gyroscope, GPS), transmitted to a FastAPI backend with SQLite storage. Machine learning models (Random Forest, Gradient Boosting) analyze the integrated dataset to generate health scores and RUL predictions. An optimization layer using Linear Programming generates cost-effective maintenance schedules for fleet operators. A shared React Native mobile application provides the user interface across all platform components. Expected outcomes include reduced unexpected breakdowns, lower maintenance costs through optimized scheduling, and improved vehicle reliability. The system aligns with SDG 9 (Industry, Innovation, Infrastructure) and SDG 11 (Sustainable Cities and Communities). Additionally, the system connects recommended maintenance actions to an integrated marketplace for compatible parts and garage/service booking.

Keywords: *Predictive maintenance, Remaining Useful Life, OBD-II, machine learning, driver behavior analysis*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
DSR	Design Science Research
GPS	Global Positioning System
IoT	Internet of Things
LP	Linear Programming
MAE	Mean Absolute Error
ML	Machine Learning

OBD-II	On-Board Diagnostics (Second Generation)
PDPA	Personal Data Protection Act
RMSE	Root Mean Squared Error
RUL	Remaining Useful Life
MCDA	Multi-Criteria Decision Analysis
ELT	Estimated Lead Time

1. INTRODUCTION

1.1 Background and Context

Vehicle maintenance is a critical aspect of transportation safety and efficiency. Most vehicle owners in Sri Lanka follow either reactive maintenance (repairs after failure) or time-based scheduled maintenance (fixed intervals). Both approaches have significant limitations. Reactive maintenance leads to unexpected breakdowns, accidents, and expensive emergency repairs mechanical failures account for approximately 10 - 15% of road accidents globally [1]. In Sri Lanka, where many vehicles are older and road infrastructure is constrained, a single stalled vehicle on a two lane road can reduce capacity by 50% [6].

Scheduled maintenance does not account for actual vehicle usage. A vehicle driven aggressively on rough roads wears its components faster than one driven gently, yet both receive the same service interval. Research shows that aggressive driving can accelerate component degradation by 20 - 40% [4]. This results in some components being replaced unnecessarily while others fail before their scheduled service date.

Predictive maintenance uses real-time data from the vehicle and its usage patterns to estimate when a component is likely to fail, enabling timely intervention. Industry reports indicate that predictive maintenance can reduce unscheduled downtime by 30 - 40% and cut overall costs by approximately 20% [1][7].

This research proposes a Predictive Vehicle Maintenance System that integrates three data sources real-time vehicle telemetry from OBD-II sensors, driver behavior captured through smartphone sensors, and historical service record to estimate the health of four critical components (tires, brake pads, engine oil, batteries) and predict their Remaining Useful Life (RUL). Since standard OBD-II does not provide direct readings for brake pad thickness, tire tread, or oil quality, the system infers degradation from indirect signals combined with behavior data and service history. A shared React Native mobile application serves as the unified interface for the broader Kaduna.lk platform.

This project is part of the Kaduna.lk platform—a unified smart platform for roadside assistance and automotive services in Sri Lanka, addressing dispatch optimization, geo-intelligence, predictive maintenance, and compliance. This proposal focuses on the Predictive Maintenance

component. The research aligns with SDG 9 (Industry, Innovation, Infrastructure) and SDG 11 (Sustainable Cities and Communities), and falls under the IoT and Smart Transportation Systems research cluster.

1.2 Literature Review

A comprehensive review was conducted across academic databases (IEEE, Springer, MDPI) and industry reports to understand the current state of predictive vehicle maintenance research.

1.2.1 AI-Driven Predictive Maintenance

Arena, F., et al. (2022) reviewed AI-driven predictive maintenance in the automotive domain [1], mapping how OBD-II sensor data and machine learning models predict component failures and estimate RUL. The study reports 30–40% reduction in unscheduled downtime. However, most reviewed studies focus exclusively on vehicle sensor data without considering driver behavior or service history.

1.2.2 OBD-II Based Vehicle Diagnostics

Michailidis, A., et al. (2023) examined ML applications for fault detection using OBD-II data [2], achieving > 85% detection accuracy for engine faults using ensemble methods. However, the approach is limited to engine-related diagnostics and does not consider driver behavior or service context. The study acknowledges that OBD-II alone provides limited visibility into components like tires and brake pads.

1.2.3 IoT-Based Vehicle Monitoring

Kumar, P., and Singh, A. (2023) developed an IoT vehicle health monitoring system using ELM327 adapters connected via Bluetooth to a mobile app [3]. While demonstrating the feasibility of affordable aftermarket monitoring, the system provides only real-time display without predictive capabilities or ML integration.

1.2.4 Driver Behavior Analysis

Smith, J., and Kumar, R. (2024) demonstrated that driving style significantly impacts vehicle wear [4]. Using smartphone sensors, they found aggressive drivers experienced 25–35% higher brake pad wear and 20–30% faster tire degradation. However, the study does not integrate vehicle sensor data or generate RUL predictions.

1.2.5 Maintenance Optimization

Zhang, L., et al. (2023) investigated optimization-based maintenance scheduling using MILP [5], demonstrating 15–25% cost reduction versus fixed-interval scheduling. However, the optimization relied on predetermined degradation rates rather than real-time sensor data.

1.2.6 Fleet Telematics

Volpis (2025) documented industry adoption of telematics for predictive fleet maintenance [7]. While showing measurable fleet uptime improvements, commercial solutions generally analyze vehicle telemetry or driver behavior separately, not in combination.

Source	Approach	Key Finding	Limitation
[1]	AI/IoT literature review	PdM reduces downtime 30–40%	No driver behavior or service data
[2]	ML fault detection (OBD-II)	>85% engine fault accuracy	Engine only; no behavior context
[3]	IoT monitoring (ELM327)	Feasible budget monitoring	Display only; no prediction
[4]	Driver behavior (smartphone)	25–35% faster brake wear	No sensor data or RUL prediction
[5]	MILP scheduling	15–25% cost reduction	No real-time data integration
[7]	Fleet telematics guide	Industry adoption growing	Single source; developed markets

Table 1: Comparison of Related Work

1.3 Research Gap

Despite extensive research in predictive vehicle maintenance, the literature review reveals several critical gaps that the proposed system addresses:

Gap 1: Isolated Data Sources.

Existing predictive maintenance systems typically rely on a single data source. OBD-II based approaches [1][2] analyze vehicle telemetry but ignore driver behavior and service history. Driver behavior studies [4] quantify driving style impact but lack vehicle sensor integration.

Fleet telematics solutions [7] monitor vehicle health but rarely combine all three data sources. No published work integrates vehicle telemetry, driver behavior, and historical service records into a unified predictive framework for RUL estimation.

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Gap 2: Prediction Without Actionable Scheduling.

Most ML-based predictive maintenance research stops at failure prediction or RUL estimation [1][2]. The output is a probability or estimated time to failure, but the system does not generate a recommended action plan. Vehicle owners receive a warning but no guidance on when to schedule service, how to prioritize multiple components approaching failure, or how to minimize total maintenance cost. The gap between prediction and actionable maintenance scheduling remains largely unaddressed.

Gap 3: Limited Contextual Awareness.

Current models treat component degradation as independent of usage context. A tire is predicted to fail based on its sensor readings alone, without considering that the driver brakes aggressively (accelerating brake pad wear), frequently overloads the vehicle (increasing tire stress), or has a history of delayed maintenance (compounding degradation). The absence of contextual factors—driving style, usage intensity, and maintenance history limits prediction accuracy and practical relevance.

Gap 4: Limited Applicability to Developing Markets.

Most predictive maintenance solutions are designed for modern vehicle fleets in developed markets, where vehicles have factory-installed advanced sensors and data infrastructure is well established [7]. In Sri Lanka, many vehicles are older models without built-in telematics, roads vary significantly in quality, and professional maintenance practices are inconsistent. There is no published predictive maintenance framework specifically adapted for these conditions—using affordable aftermarket OBD-II adapters and smartphone sensors as the primary data sources.

Gap 5: No Optimization-Driven Maintenance Planning.

While optimization-based scheduling has been explored independently [5], it has not been connected to real-time sensor data and ML-based RUL predictions. Existing optimization approaches use predetermined degradation curves rather than adapting to actual vehicle

condition. Connecting ML prediction outputs directly to an optimization engine that generates cost-minimized, constraint-aware maintenance schedules represents a significant unexplored opportunity.

Gap 6: Disconnection from Service and Parts Ecosystems.

Current research focuses heavily on detecting a vehicle fault but ignores the actual process of repairing it. Even when an AI correctly warns a driver of an upcoming failure, there is no direct link to a marketplace for spare parts or a directory of verified garages. This creates a "dead end" in the user experience. Without a built-in marketplace, the driver must manually search for parts and compare service prices, which often leads to confusion and maintenance delays. There is currently no integrated system that provides a seamless transition from an AI-calculated health alert to a confirmed service booking.

The proposed research addresses these gaps by integrating three data sources (vehicle telemetry, driver behavior, service history) to estimate the Remaining Useful Life (RUL) of critical components using machine learning. Unlike existing models, this framework creates a seamless transition to repair by connecting diagnostic outputs directly to an integrated service marketplace. This multi-source, optimization-driven, and market-aware approach represents the unique contribution of this research.

2. OBJECTIVES

2.1 Main Objective

To develop a predictive vehicle maintenance system that integrates real-time vehicle telemetry, driver behavior data, and historical service records to estimate the Remaining Useful Life (RUL) of critical vehicle components and generate cost-optimized maintenance schedules, and connect predicted maintenance needs to a parts marketplace and garage/service booking workflow.

2.2 Specific Objectives

SO1: To collect and integrate multi-source data including real-time vehicle telemetry (OBD-II via ELM327), driver behavior metrics (smartphone sensors), and historical service records into a unified dataset for predictive maintenance modeling.

SO2: To develop composite component health indicators that quantify the condition and wear level of four critical vehicle parts: tires, brake pads, engine oil, and battery, using a weighted combination of telemetry, behavior, and service data.

SO3: To predict the Remaining Useful Life (RUL) of each monitored component using machine learning models (Random Forest and Gradient Boosting), targeting a prediction accuracy of MAE < 15% of actual component lifespan.

SO4: To implement an optimization-based maintenance scheduling framework using Linear Programming that generates maintenance schedules minimizing total maintenance cost while respecting component health thresholds and vehicle availability constraints.

SO5: To provide actionable maintenance recommendations through a user dashboard and marketplace integration that prioritizes components based on predicted RUL, urgency, and cost, and suggests compatible parts and recommended garages for timely service booking.

SO6: To evaluate the system's performance through simulation experiments, assessing RUL prediction accuracy (MAE, RMSE), maintenance cost reduction compared to fixed-interval scheduling, and reduction in simulated unexpected breakdowns.

Objective	Key Deliverables	Evaluation Method
SO1 – Data Collection & Integration	Combined dataset from OBD-II telemetry, driver behavior data, and service records	Check data completeness and successful integration of different data sources
SO2 – Component Health Indicators	Health score calculation for tires, brake pads, engine oil, and battery	Compare calculated health scores with expected wear levels
SO3 – RUL Prediction Model	Machine learning models (Random Forest, Gradient Boosting) predicting Remaining Useful Life	Measure prediction accuracy using **MAE and RMSE**
SO4 – Maintenance Scheduling Optimization	Algorithm that generates cost-efficient maintenance schedules	Compare optimized schedule cost with fixed maintenance schedule
SO5 – Maintenance Recommendation Dashboard	Simple dashboard showing component health, RUL, marketplace suggestions, the and maintenance alerts	User testing to check clarity and usefulness of recommendations
SO6 – System Performance	Final system prototype and simulation	Evaluate prediction accuracy, cost

Evaluation	results	savings, and reduction of simulated breakdowns

Table 2: Mapping of Objectives to Deliverables

3. METHODOLOGY

3.1 Research Approach

This study adopts a Design Science Research (DSR) approach combined with experimental methodology. The focus is on designing and evaluating a software artifact the predictive vehicle maintenance system that provides actionable maintenance recommendations based on integrated multi-source vehicle data. The experimental component involves testing the system under simulated conditions to evaluate prediction accuracy and optimization effectiveness.

Research Questions:

- RQ1: Can ML models trained on integrated OBD-II, driver behavior, and service data predict RUL more accurately than single-source models?
- RQ2: Does optimization-based scheduling reduce maintenance cost and breakdowns compared to fixed-interval scheduling?
- RQ3: How does driver behavior quantitatively affect component degradation rates?

Hypotheses:

- H1: Multi-source models achieve at least 20% lower MAE than single-source models.
- H2: Optimization-based scheduling reduces total maintenance cost by at least 15%.
- H3: Driver behavior scores correlate significantly (Pearson $r > 0.5$) with degradation rates.

3.2 Data Collection and Preprocessing

Vehicle Telemetry (OBD-II)

An ELM327 Bluetooth OBD-II adapter reads standardized PIDs at 5–10 second intervals during driving: engine RPM, engine load, coolant temperature, battery voltage, throttle position, vehicle speed, and DTCs. The React Native app communicates with the adapter via Bluetooth and buffers data locally during trips. After each trip, summary statistics and key raw data points are uploaded in batch to the FastAPI backend.

Standard OBD-II does not provide direct readings for brake pad thickness, tire tread, or oil quality. The system adopts an inference-based approach: degradation is estimated from indirect OBD-II signals (braking patterns from speed data, engine load profiles, temperature trends) combined with driver behavior metrics and service history. Battery voltage is the one directly readable component parameter.

Driver Behavior (Smartphone Sensors)

The app captures driving behavior using accelerometer (braking/acceleration intensity), gyroscope (sudden maneuvers), and GPS (speed, distance, route). Derived metrics include: braking frequency, cornering aggressiveness, speed variability, and an overall behavior classification (calm/moderate/aggressive).

Historical Service Records

Users input maintenance history including past service dates, component replacements, and mileage. Service records provide context on component age since last replacement and maintenance consistency.

The datasets required to train the machine learning models will be sourced from open-source repositories such as Kaggle, covering both OBD-II diagnostics and driver behavior telemetry. Additionally, primary data will be gathered through manual methods, including semi-structured interviews, questionnaires, and the collection of physical service records from approximately 20 vehicle owners. This combination of synthetic, open-source, and real-world historical data ensures a comprehensive foundation for training the RUL (Remaining Useful Life) and Diagnostic Triage models.

Component	OBD-II Signals (Indirect)	Behavior Metrics	Service Data
Tires (Inferred)	Speed, throttle, distance	Cornering aggression, acceleration patterns	Last replacement, km since replacement
Brake Pads (Inferred)	Deceleration patterns from speed data	Braking frequency, braking intensity	Last replacement, km since replacement
Engine Oil (Inferred)	Engine RPM, load %, coolant temperature	High-RPM driving, stop-and-go ratio	Last oil change, km since change

Table 3 : Component Health Indicator Parameters

3.3 Feature Engineering and Health Indicators

A composite health score (0–100) is calculated for each component using a weighted formula:

$$Health_Score = w_1 \times Telemetry_Score + w_2 \times Behavior_Factor + w_3 \times Service_Factor$$

Initial weights (w_1=0.45, w_2=0.30, w_3=0.25) are assigned based on domain knowledge and adjusted through cross-validation. The telemetry score captures current condition from OBD-II signals, the behavior factor reflects driving style impact on degradation, and the service factor accounts for maintenance history and time since last service.

3.4 RUL Prediction Models

Two ML algorithms are implemented and compared: Random Forest Regressor (robust to noise, provides feature importance) and Gradient Boosting Regressor (higher accuracy on structured data, implemented via XGBoost/LightGBM). Both are trained on the integrated feature set with 80/20 train-test split and 5-fold cross-validation. A comparison with single-source models (telemetry only, behavior only) quantifies the improvement from multi-source integration.

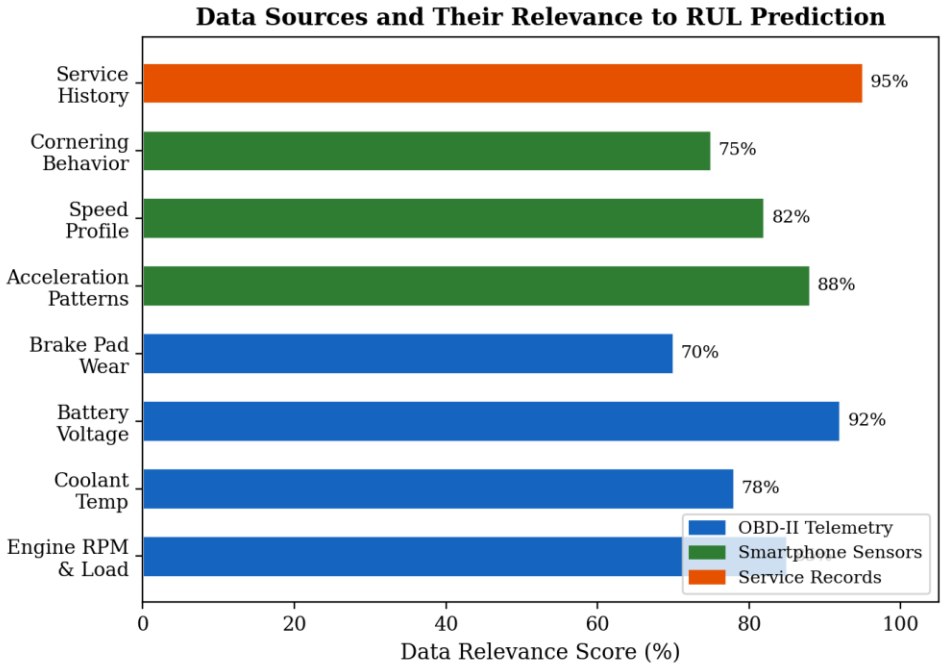


Figure 1:Data Sources and Their Relevance to RUL Prediction

3.5 Maintenance Scheduling

The system uses a two-tier scheduling approach. For individual vehicle owners, a rule-based priority scoring ranks components by urgency (predicted RUL, health score, replacement cost) essentially "which component needs attention first." For fleet managers with multiple vehicles, LP/MILP optimization minimizes total fleet maintenance cost subject to constraints: component health thresholds, vehicle availability, service center capacity, and budget limits. The optimization is implemented using PuLP.

Garage suggestion: The system ranks nearby service centers using a Multi-Criteria Decision Analysis (MCDA) framework integrated with PuLP-based linear programming. This optimization engine evaluates service capability, real-time availability, and distance to select the best repair option by minimizing a total cost function that includes estimated labor and travel overhead. By processing operational constraints such as garage-specific technical certifications and current bay capacity, the system moves beyond simple proximity to provide a mathematically optimized "Best-Fit" repair solution. [14][13][15]

Parts suggestion: For each component approaching replacement, the system executes a Compatibility Mapping Algorithm that searches for compatible parts using vehicle-specific requirements and VIN-level compatibility rules. It retrieves structured data on price, real-time availability, and Estimated Lead Time (ELT) from partner suppliers to ensure a seamless transition from diagnosis to repair. By prioritizing parts based on a "Value-to-Durability" ratio, the engine ensures that replacement components are matched to the predicted remaining life of interconnected vehicle systems, preventing redundant labor costs. [13][15]

3.6 Validation Strategy

Metric	Description	Target
MAE (RUL)	Mean Absolute Error of RUL predictions	< 15% of actual lifespan
RMSE (RUL)	Root Mean Squared Error	< 20% of actual lifespan
Cost Reduction	vs. fixed-interval scheduling	≥ 15% reduction
Breakdown Reduction	Simulated unexpected failures prevented	≥ 25% reduction
Multi-source Gain	MAE improvement: multi vs. single source	≥ 20% improvement

Table 4 : Evaluation Metrics and Targets

Validation uses simulation-based comparison. Baseline 1 (fixed-interval) uses manufacturer recommended schedules (e.g., oil change every 5,000 km). Baseline 2 (reactive) replaces components only at failure. A simulated fleet with varying driving patterns is run through all three strategies over 12 months, comparing breakdowns, costs, and unnecessary replacements.

3.7 Tools and Technologies

Category	Tool	Purpose	Justification
Data Collection	ELM327 OBD-II Adapter	Vehicle telemetry	Affordable Bluetooth adapter; standard OBD-II
Data Collection	Smartphone Sensors	Driver behavior	Built-in accelerometer, gyroscope, GPS
Backend	FastAPI (Python)	REST API	Lightweight; same language as ML pipeline
Database	SQLite	Data storage	Serverless; zero-config; ideal for prototype
ML	Scikit-learn, Pandas, NumPy	RUL prediction	Industry-standard ML libraries
Optimization	PuLP (Python)	Schedule optimization	Open-source LP/MILP solver
Mobile App	React Native	Shared platform app	Cross-platform; shared across 4 team members
Version Control	Git / GitHub	Code management	Team collaboration and version tracking

Table 5 : Tools and Technologies

3.8 Ethical Considerations

Data is collected from volunteer vehicles with informed consent. All personal identifiers are anonymized. The system follows privacy-by-design principles compliant with PDPA (Sri Lanka) and GDPR. GPS coordinates are hashed, and only minimum necessary data is collected and retained. Participants can withdraw and request data deletion at any time.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed Predictive Vehicle Maintenance System follows a modular, layered architecture designed for scalability and maintainability. The system comprises five primary layers, each with distinct responsibilities: Data Acquisition, Data Processing and Analytics, Prediction and Optimization, Backend and Data Storage, and User Interface and Monitoring. This separation allows independent development and testing of each layer while maintaining clean interfaces between them.

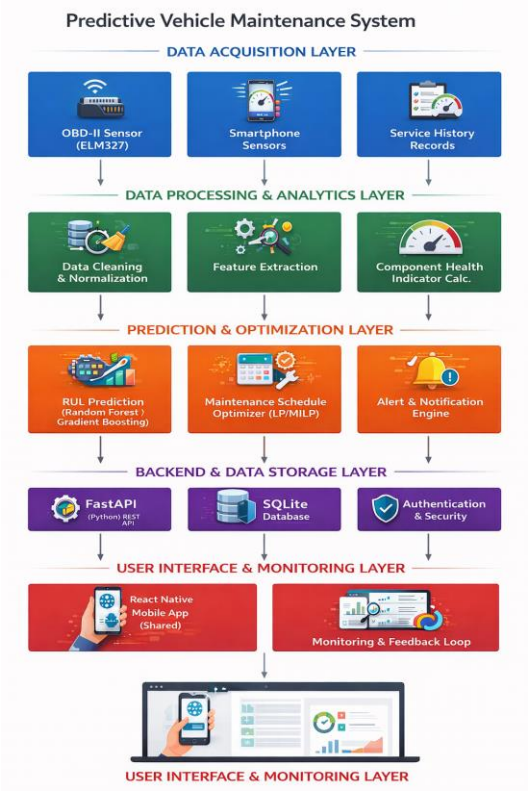


Figure 2 : High-Level System Architecture

4.2 Component Interactions and Data Flow

Data flows through the system in a structured pipeline. OBD-II sensors and smartphone sensors continuously capture vehicle telemetry and driver behavior, transmitting data to the backend via the mobile application. Historical service records are stored in the database and merged with live data during preprocessing. The integrated dataset feeds into the Component Health Calculator, which generates health scores for each monitored component. These scores flow into the RUL

Prediction Module for lifespan estimation. Predicted RUL values, along with component costs and vehicle constraints, are passed to the Optimization Engine to produce a recommended maintenance schedule. Results are exposed via the RESTful API to the Dashboard, where users can visualize insights and receive alerts. Actual maintenance outcomes are fed back into the system for continuous model refinement.

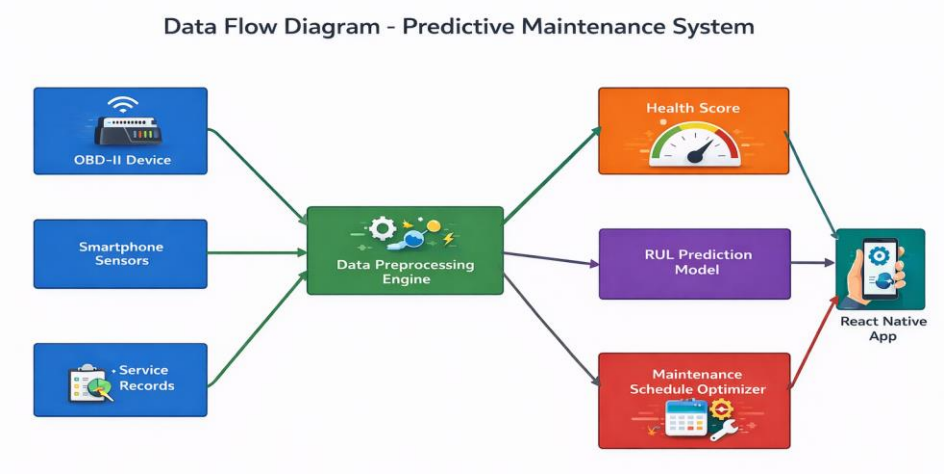


Figure 3 : Data Flow Diagram

4.3 Scalability and Security

The modular design supports addition of new vehicles, components, and data sources. The Python backend with SQLite can migrate to production-grade databases as the user base grows. Security includes role-based access control, HTTPS/TLS encryption, data anonymization, and PDPA-compliant storage practices.

4.4 Marketplace & Garage Recommendation Engine

The workflow links RUL predictions to parts availability and garage/service booking for proactive maintenance.

The system queries the marketplace for compatible parts and retrieves supplier information (price, stock/availability, and lead time). For garages, it uses partner profiles to estimate service capability and estimated turnaround time. An optimization step (LP/MILP) selects the repair option that minimizes total expected cost under constraints such as availability and service capacity. [5][14][13][15]

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role in the System
Vehicle Owners / Drivers	Monitor vehicle health, receive maintenance alerts, and understand how driving behavior affects vehicle condition.
Fleet Managers	Monitor the health of multiple vehicles, plan maintenance schedules, reduce downtime, and optimize maintenance costs.
Service Technicians	Access predicted maintenance schedules and component health data to perform preventive repairs efficiently.
System Administrators	Manage system configurations, user accounts, and ensure data security and proper system operation.

Table 6 : Stakeholder Identification

5.2 Functional Requirements

ID	Requirement
FR-01	Collect vehicle telemetry from OBD-II sensors (engine diagnostics, battery voltage, speed).
FR-02	Capture driver behavior (acceleration, braking, cornering) via smartphone sensors.
FR-03	Store and manage historical service records and maintenance logs.
FR-04	Preprocess and integrate telemetry, behavior, and service data into a unified dataset.
FR-05	Calculate composite health scores (0–100) for each monitored component.
FR-06	Predict Remaining Useful Life (RUL) of tires, brake pads, engine oil, and battery.
FR-07	Generate optimized maintenance schedules (rule-based for individuals, LP for fleets).
FR-08	Provide push notification alerts when components approach critical thresholds.
FR-09	Display health scores, RUL predictions, and recommendations on the mobile app.
FR-10	Classify driving behavior as calm, moderate, or aggressive with wear impact display.
FR-11	Allow users to manage vehicle profiles, service records, and predicted schedules.
FR-12	Map predicted component failures to compatible parts and verified service providers.

Table 7 : Functional Requirements

5.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Process data and update health scores within 5 seconds.
NFR-02	Performance	App loads health information within 3 seconds.
NFR-03	Security	All data encrypted via HTTPS/TLS; PDPA-compliant storage.
NFR-04	Security	Role-based authentication and data anonymization.
NFR-05	Usability	Intuitive charts and color-coded health indicators.
NFR-06	Reliability	No loss of critical telemetry or maintenance data.
NFR-07	Scalability	Support addition of new vehicle types and fleet scaling.

Table 8 : Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market:

Daily vehicle owners and commuters in Sri Lanka, commercial fleet operators (logistics, delivery, ride-hailing services), vehicle service centers, and insurance companies interested in reducing vehicle breakdown risks. This module complements the Kaduna.lk dispatch platform by enabling proactive vehicle health monitoring.

6.2 Value Proposition:

Adds a predictive intelligence layer to the Kaduna.lk roadside assistance platform. By analyzing vehicle telemetry, driver behavior, and service history, the system predicts component failures before they occur. This reduces unexpected breakdowns, lowers maintenance costs through optimized service scheduling, and improves road safety while enabling more efficient dispatch operations. It also connects recommended maintenance actions to an integrated marketplace for parts and garage/service booking, reducing repair lead time.

6.3 Revenue Model:

The predictive maintenance module is integrated into the Kaduna.lk SaaS platform. Individual users access basic vehicle health monitoring within the freemium tier, while premium subscriptions unlock advanced predictive analytics, maintenance alerts, and service recommendations. Fleet operators and logistics companies access fleet-level predictive analytics and maintenance optimization through enterprise subscriptions and API integration. Marketplace operations can generate additional revenue through referral commissions and booking fees from partner garages and suppliers.

6.4 Competitive Advantage:

- Integrated platform combining predictive maintenance and Smooth transmission for marketplace.
- Multi-source data analysis using OBD-II telemetry, driver behavior data, and service records.
- Optimization-based maintenance scheduling to reduce downtime and service costs.
- Affordable implementation using standard OBD-II adapters and smartphones.

6.5 IP Considerations:

The predictive maintenance framework, including the multi-factor health scoring model, Remaining Useful Life prediction algorithms, and maintenance scheduling optimization approach, may be eligible for intellectual property protection. The research methodology and results may be published academically while the production implementation remains proprietary within the Kaduna.lk platform.

7. BUDGET AND JUSTIFICATION

Category	Item	Cost (LKR)	Justification
Hardware	ELM327 OBD-II Adapter (x2)	5,000–10,000	Bluetooth adapters for test vehicles
Hardware	Development laptops (existing)	0	Team members use personal laptops
Software	Python, FastAPI, React Native	0	Open-source frameworks
Cloud	Cloud hosting (Render/Vercel)	0–15,000	Free tier for dev; paid for production
Data	Public automotive datasets	0	Kaggle datasets freely available
Human Effort	4 members × 10 months	0 (academic)	Final-year research project
Printing	Thesis printing and binding	10,000–15,000	Final submission copies
Misc.	Internet, transport, stationery	15,000–20,000	Ongoing expenses
	TOTAL ESTIMATED	30,000–60,000	

Table 9 : Budget Breakdown

8. WORK BREAKDOWN STRUCTURE (WBS)

WP	Task	Responsible	Timeline	Deliverables
WP1	Literature Review & Data Collection	Herath	Month 1–2	Literature survey, datasets, collection plan
WP2	Data Preprocessing	Herath	Month 2–4	Cleaned, merged dataset; preprocessing scripts
WP3	Feature Engineering & Health Indicators	Herath	Month 3–5	Feature set, health score formulas
WP4	RUL Model Development	Herath	Month 4–7	Trained RF/GB models, evaluation results
WP5	Maintenance Scheduling	Herath	Month 5–7	Rule-based + LP/MILP implementation
WP6	App Screens & API	Herath +	Month 6–8	React Native screens, FastAPI endpoints

		All members		
WP7	Integration & Testing	All members	Month 7–9	Integrated system, end-to-end testing
WP8	Evaluation & Writing	Herath	Month 8–10	Evaluation report, thesis, presentation

Table 10 : Work Breakdown Structure

9. GANTT CHART

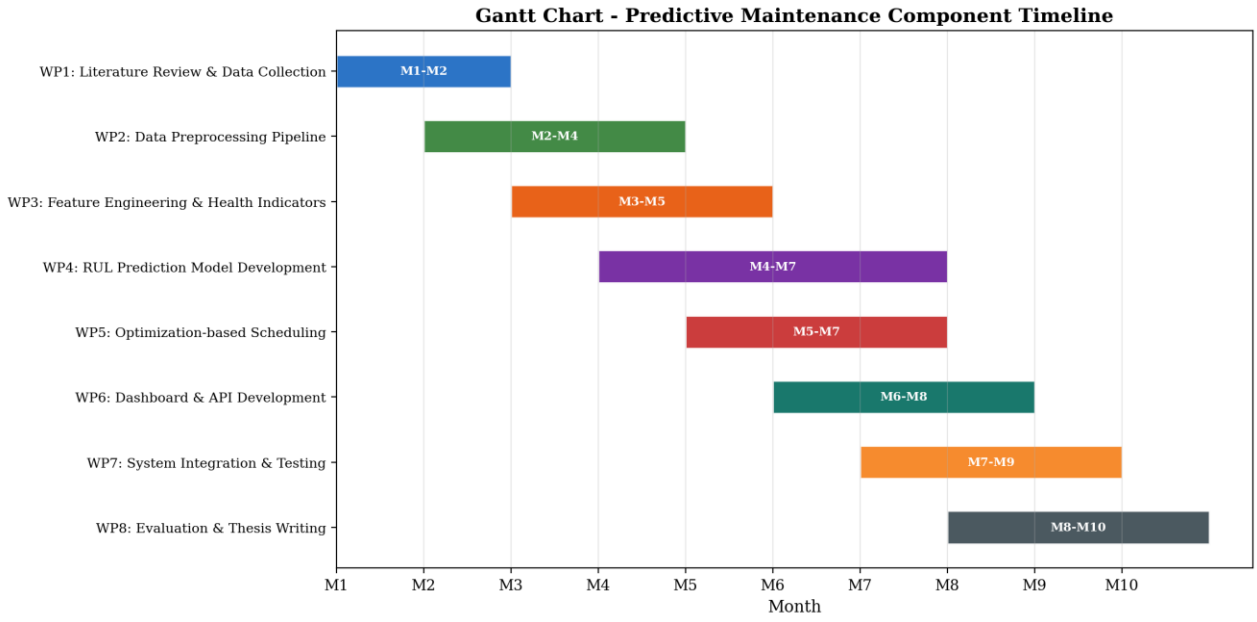


Figure 4 : Gantt Chart

Key Milestones:

- Month 2: Literature review complete; initial datasets collected
- Month 4: First RUL prediction model trained and evaluated
- Month 6: Model comparison complete; scheduling implemented
- Month 8: App screens deployed; system integrated
- Month 10: Final evaluation, thesis, and presentation submitted

Additional WP: Marketplace integration & garage booking workflow (parts availability, garage ranking, booking notifications, and feedback loop).

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APPENDICES

Appendix A: Component Health Score Calculation

$$Health_Score_i = w_1 \times T_i + w_2 \times B_i + w_3 \times S_i$$

Where T_i = normalized telemetry score, B_i = behavior impact factor, S_i = service history factor. Weights: w_1=0.45, w_2=0.30, w_3=0.25 (adjusted via cross-validation). Battery health uses direct voltage: T_battery = normalize(voltage, min_v, max_v). For inferred components (tires, brakes, oil), T_i is derived from proxy signal combinations.

Appendix B: Risk Analysis Matrix

Risk	Likelihood	Impact	Mitigation
Insufficient real vehicle data	High	High	Use Kaggle datasets + synthetic data; validate with limited real data
OBD-II adapter incompatibility	Medium	Medium	Test multiple ELM327 models; use only standard PIDs
ML model overfitting	Medium	High	Cross-validation, regularization, model comparison
Integration delays	Low	Medium	Design API independently; mock other components
Sensor noise in behavior data	Medium	Medium	Signal filtering (low-pass, Kalman); calibration
Scope creep	Medium	High	Strict focus on 4 components and defined objectives
Privacy concerns	Low	High	Informed consent; anonymization; PDPA compliance

Table 11 : Risk Analysis Matrix

Appendix C: UI Mockups

